CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

CAPSTONE PROJECT MASTER GUIDE

Complete Student Project Pack

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
Group Members	Freek Schaefer-Gerber, Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Final Submission
Submission Date	29 October 2026

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1. Purpose of the Capstone Project

Develop and evaluate a working software solution that addresses a clearly defined real-world problem. The project must demonstrate research, requirements analysis, design, development, version control, testing, evaluation and professional presentation.

Develop and evaluate a working software solution that addresses a clearly defined real-world problem. This project pack documents the complete lifecycle of SmartDesk, from initial research and requirements gathering through to development, testing, and final evaluation.

2. Six-Chapter Structure

Chapter 1: Introduction; Chapter 2: Literature Review; Chapter 3: Requirements & Methodology; Chapter 4: System Design & Development; Chapter 5: Testing, Results & Evaluation; Chapter 6: Conclusion & Recommendations.

- Chapter 1: Introduction
- Chapter 2: Literature Review
- Chapter 3: Requirements & Methodology
- Chapter 4: System Design & Development
- Chapter 5: Testing, Results & Evaluation
- Chapter 6: Conclusion & Recommendations

3. Minimum Evidence Required

Approved topic; research problem, aim, objectives and questions; literature review and research gap; functional and non-functional requirements; methodology; UML; database design where applicable; source code; Git evidence; testing; evaluation; final dissertation; presentation and live demonstration.

- Research & Planning: Approved topic, problem statement, aims, objectives, research questions, and literature review with identified gap.
- Design & Development: Functional/non-functional requirements (SRS), methodology, UML diagrams, ERD (database design), and complete Python/Flask source code.
- Testing & Evaluation: Automated `pytest` suite, 20 manual test cases, Defect Log, Requirements Traceability Matrix (RTM), and Chapter 5/6 evaluations.
- Project Management & Presentation: Project Charter, Risk Register, Git commit evidence, final presentation slides, and demo script.

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4. File Naming Rule

Use clear names such as GroupName_Chapter1_v1.docx. Do not submit files named final-final2 or document1. Keep evidence in the matching folder.

All project documentation has been logically organised into numbered folders (00 to 12) to match the project phases. Source code is maintained in the GitHub repository, and all evidence (screenshots, test logs) is filed within its respective folder.

5. Submission Quality

Use professional academic writing, consistent formatting, page numbers, references, labelled figures/tables, readable diagrams, and evidence that every member contributed.

The final dissertation uses professional academic writing, consistent formatting, automatic Table of Contents, page numbers, Harvard/IEEE referencing, and labelled figures/tables. Git commit history and the Individual Contribution Form provide evidence that both members contributed equally to the project.

6. Final Check

Use the assessment checklist and marking rubric before submission. The application must be demonstration-ready.

The assessment checklist and marking rubric have been reviewed. The final application is fully functional, secured against critical vulnerabilities, and the demonstration environment is verified and ready for the live presentation on 29 October.